

Project Euler

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Problem 1: Multiples of 3 or 5

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Problem 1



If we list all the natural numbers below 10 that are multiples of 3 or 5, we get 3, 5, 6 and 9. The sum of these multiples is 23.
Find the sum of all the multiples of 3 or 5 below 1000.

Maths

$$\text{Answer} = \sum 3x + \sum 5x - \sum 15x$$

$$\sum 3x = 3(1 + 2 + 3 + \dots + 333)$$

$$\sum 5x = 5(1 + 2 + 3 + \dots + 199)$$

$$\sum 15x = 15(1 + 2 + 3 + \dots + 66)$$

$$\text{Summation Formula } \frac{(n)}{2}(n + 1)$$

$$= 3 \left(\frac{333(333 + 1)}{2} \right) + 5 \left(\frac{199(199 + 1)}{2} \right) - 15 \left(\frac{66(66 + 1)}{2} \right)$$

$$= 3 \left(\frac{333 \cdot 334}{2} \right) + 5 \left(\frac{199 \cdot 200}{2} \right) - 15 \left(\frac{66 \cdot 67}{2} \right)$$

$$= 233168$$

Code

```
1 total = 0
2
3 for number in range(1, 1000):
4     if number % 3 == 0 or number % 5 == 0:
5         total += number
6
7 print(total)
```

233168



Congratulations, the answer you gave to problem 1 is correct.